§19 of the isogeny-fibers note: rounds 5 and 6, verdicts and corrections

Date 2026-10-24. **Type** documentation of existing results; no experiment was run for this report and no number below is newly computed unless marked *derived*. **Branch** `research/note-s19-rounds-5-6`, base `origin/main` `5cb553810e`. Toy and synthetic instances only; no sub-rho claim and no claim about a deployed curve.

Question. What did the fifteen leads of rounds 5 and 6 (R5.1–R5.4, R6.1–R6.11) find, against their preregistered hypotheses and falsifiers, and which earlier statements of `RESEARCH_ISOGENY_FIBERS_ECDLP.md` must change in light of them and of the R6.10 literature check?

Method. Read each lead's `RESULTS.md` on `main` (R6.2: its run report and the raw `batch.json` and `paths.json`), and for the five pull requests still open on this date (R5.3 2192, R6.1 2511, R6.4 2240, R6.8 2196, R6.9 2502) the `RESULTS.md` on the pull request's branch, together with the owner's reviews and the status notes in the pull request descriptions. Verdicts and headline numbers are transcribed; intervals are the sources'. Source heads: R5.3 `e8e0bf1296`, R6.8 `7d6d50d268`, R6.4 `325c8345b8`, R6.9 `6ac9a3877e`, R6.1 `ff6fd1dc6c`.

The verdict figure (Figure 1) is on the last page (landscape).

Findings, including negative and unresolved ones

- **Falsifier fired.** R6.5: the A-pair correction does not explain the recorded single-base value 1.02 (mean residual −0.030 before, −0.085 after correction); the correction itself is confirmed in a 200-base ensemble (1.130, 1.059, 1.033 at $|F| = 36, 72, 144$). R5.3 (unmerged): "exponent drops toward 0.5–0.6" refuted, 0.798 [0.775, 0.820]. R6.4 (unmerged): ascending against descending differs for the ψ route (0.501 [0.457, 0.548]) and the Φ route (0.360 [0.347, 0.373]). R6.9 (unmerged): relation rate off the $1/g!$ heuristic by more than 10× (the source says 6 of 34 runs; its own table has 7).
- **Not fired.** R5.1, R5.4, R6.3, R6.6, R6.7 (read against the PROGRAM falsifier), R6.11, R6.1 and the R6.8 census.
- **Open.** R5.2 as a cost statement; R6.2 (no paired interval, two known defects: planner applicability and an asymmetric-edge path search; path numbers pending correction); R6.8′s exact-simulation theorem (proof claim withdrawn in the pull request); the generic-kernel exponent of R6.4 (1.919 [1.724, 2.115], insufficient evidence either way, restricted cost model).
- **Prior work.** R6.10: the kernel-saturation penalty (the μ^{m-2} clause of Prop. 15.2) is anticipated by Faugère–Huot–Joux–Renault–Vitse, EUROCRYPT 2014, §3.2.
- **Source defect found while reading.** R6.7′s merged `RESULTS.md` says 12 of 18 per-curve intervals contain 1; its records give 11, 9 and 11 of 18 (unpriced, priced as run, priced essential). The note uses the record counts. The merged PDF was not regenerated here (frozen evidence).

What changed in the note (research note, not a result)

New §19 (one subsection per lead, synthesis table, settled/open list); header status line; §0 items 3, 5, 8, 9; §6 H6 and §13.4 genus-2 attribution; FHJRV citation in §14.1, §15.2, §17.4, §17.7; §15.2 erratum from R6.5; §15.3 xedni and Koshelev wording; §17.2 patent sentence; §17.7 walk qualifier; §18 replaced by the R6.10 list. Measured values in earlier sections are unchanged.

Graphs checked and left unchanged

`docs/index-calculus-scoreboard.html`, `docs/ic/progress-timeline.json`, `docs/performance-gains/`, `docs/ic/BOUNDARY_TARGETS.md` and `figures/` carry no isogeny-fibers round 5–6 row (a search of `origin/main` finds only generated yield-record paths in `docs/browser/data.json`, which this change does not affect). No scoreboard or ledger value changes; the study-local graphs of each lead are unchanged and are linked from §19.

Limits and unresolved checks

Verdict classes compress mixed outcomes into one column; the wording is in §19.16. Claims about open pull requests are as of the heads above. The R6.2 and R6.8 standing issues and the R6.4 and R5.3 validation holds are stated in the pull requests, not resolved here. `conductor check` was not available in this environment (no `conductor` executable), so no task was reserved; edits are confined to the note and this directory. The curve-name linter could not be run against the sparse checkout; no retired curve name is used, and the only curve slug in §19 is one already present in the note.

Commands

`venv/bin/python plot.py` (reads `verdicts.json`, writes `verdicts.svg` and `verdicts.png`); `typst.compile('report.typ')` for this PDF.



Filled marker: merged into main (2026-10-24). Hollow marker: pull request still open (UNMERGED). One verdict class per lead; see section 19.16 for the wording. Toy instances only. Source: verdicts.json.

Figure 1: Verdict class of each lead (transcribed; data `verdicts.json`, script `plot.py`). Hollow markers are unmerged pull requests, read from their branches; results there can still change.